Defer/Refer In-Class Test

4COSC006C Software Development I (IIT Sri Lanka) · 16 September 2021 · 90 minutes · 100 points · 32 questions

Question 1 (3 points)

What is used to define a block of code in Python?

- **A.** Curly braces
- **B.** Parenthesis
- **C.** Indentation
- **D.** None of the above

Question 2 (3 points)

Match the following examples to the expected result.

```
15//2
15%2
15.0/2
```

Results to choose from: 1, 7, 7.5

Question 3 (3 points)

Which error does not cause a program error, but produces incorrect results?

- **A.** runtime
- **B.** semantic
- **C.** syntax

Question 4 (3 points)

Which of the following is NOT a valid augmented assignment operator?

- **A.** *=
- **B.** -=
- **C.** x=
- **D.** +=

Question 5 (3 points)

```
x = 10
__________ (Blank 1)
```

Write the line of code that will multiply by five the value of x – ensure your answer uses an augmented assignment operator.

Question 6 (3 points)

Which of the following will read a string into variable **name**?

- **A.** name = input("Enter name ")
- **B.** String name = "Some Name"
- **C.** name = string(input("Enter name "))
- **D.** name = string("Enter name ")

Question 7 (3 points)

How would you access the substring 'end'?

```
s = 'Blended'
```

Answer: _____

Question 8 (3 points)

Which of the following is **NOT** a valid operation on a Python string variable?

- **A.** Multiplication of string. E.g., 'b' * 5
- **B.** Addition of strings. E.g., 'Good' + ' Bye'
- **C.** Changing a string. E.g., name[3] = 'a'
- **D.** Indexing a string to get a single letter of the string. E.g., module = 'maths'; module[0]

Question 9 (3 points)

```
found = True
```

Given the above, which of the following if statements includes a condition that evaluates to true?

- **A.** if found == "True":
- **B.** if found:
- **C.** All answers are correct
- **D.** if found = True:

Question 10 (4 points)

Which of the following expressions checks if a value assigned to the **number** variable falls in the range 1000 to 2000?

- **A.** number >= 1000 or number <= 2000
- **B.** number > 1000 or number < 2000
- **C.** number => 1000 or number =< 2000
- **D.** number > 1000 and number < 2000
- **E.** number => 1000 and number =< 2000
- **F.** number >= 1000 and number <= 2000

Question 11 (4 points)

To apply a discount on tickets, which of the following expressions checks whether a value for the **age** variable is either less than 16 or more than 60?

- **A.** age <= 16 and age >= 60
- **B.** age <= 16 or age >= 60
- **C.** age < 16 or age > 60
- **D.** age < 16 and age > 60
- **E.** age =<16 and age => 60
- **F.** age =< 16 or age =< 60

Question 12 (3 points)

```
y = 4
x = 5
```

Which of the following is true?

- **A.** y != 4
- **B.** not (x == 5)

C. not (y == 5)

D. not (y == 4)

Question 13 (3 points)

Which of the following will cause an error if x = 3, y = 10, and z = 0?

A. y / z

B. None will cause an error

C. y % x

D. y + 12

E. x ** z

Question 14 (2 points)

The following fragment of code is written using a for loop.

```
sum = 0
for k in range(0, 16, 3):
    sum = sum + k
print(sum)
```

The following shows the above loop re-written as a while loop. Fill in the gaps.

```
sum = 0
__________ (Blank 1)
while k < 16:
    sum = sum + k
    __________ (Blank 2)
print(sum)
```

Question 15 (3 points)

```
for i in range(10):
    print(i)
    if i == 2:
        # missing line
print('Exited loop')
```

Which keyword could be entered on the missing line to make the program jump out of the loop early so that the following is displayed?

```
0
1
2
Exited loop
```

A. exit

B. stop

C. continue

D. *(option not visible in the original screenshots)*

Question 16 (3 points)

```
colour = ["red", "green"]
fruits = ["apple", "banana", "grape"]

for x in colour:
```

```
    for y in fruits:
        print(x, y)
```

How would the above display when run?

A. red apple
green apple
red banana
green banana
red grape
green grape

B. red green apple
red green banana
red green grape

C. red
green
apple
banana
grape

D. red apple
red banana
red grape
green apple
green banana
green grape

Question 17 (3 points)

```
s = "hello"
```

Which of these is correct?

A. upper(s) returns new string with all capital letters

B. s.upper() returns new string with all capital letters

C. s.upper() returns a new string with the first letter as a capital letter

D. upper(s) returns a new string with the first letter as a capital letter

Question 18 (3 points)

Which of the following returns a sequence 1, 2, 3, 4?

A. range(1, 5)

B. None of these answers.

C. range(1, 4)

D. range(1, 2, 3, 4)

Question 19 (3 points)

Which range function call will produce the following ordered sequence: 28, 21, 14, 7

A. range(28, 7, 4)

B. range(28, 7, -7)

C. range(28, 5, -7)

D. range(7, 28, 7)

Question 20 (3 points)

Which is **true**?

- **A.** A parameter is a value passed to a function when the function is called.
- **B.** An argument is a value passed to a function when the function is called.
- **C.** An argument is the variable listed inside the parentheses in a function definition.
- **D.** A parameter is always a global variable.

Question 21 (3 points)

What is the default return value for a user-defined function that does not return any value explicitly?

- **A.** null
- **B.** An empty string
- **C.** 0
- **D.** None

Question 22 (3 points)

Consider a function named **sum**. It accepts two integer arguments and returns their sum as an integer. Fill in the blanks.

```
def sum(___________):         # Blank 1
    ans = x + y
    ___________               # Blank 2

result = ___________(4, 5)   # Blank 3
```

Question 23 (3 points)

Which of the following function headers is valid?

- **A.** def calculate(a = 2, b, c = 5):
- **B.** def calculate(a, b = 3, c = 5):
- **C.** All are invalid
- **D.** All are valid
- **E.** def calculate(a = 2, b):

Question 24 (3 points)

```
salary = 15000

def printSalary():
    salary = 16000
    print("Salary:", salary)

printSalary()
print("Salary:", salary)
```

What is the output of the following?

- **A.** Salary: 15000
  Salary: 15000
- **B.** Salary: 15000
  Salary: 16000
- **C.** Salary: 16000
  Salary: 16000
- **D.** Salary: 16000
  Salary: 15000

Question 25 (3 points)

Match the code to the expected result (give the value of **numbers** after each).

```
numbers = [1, 2, 3, 4, 5, 6, 7]
numbers.append(10)
```

```
numbers = [1, 2, 3, 4, 5, 6, 7]
numbers.insert(0, 10)
```

Question 26 (3 points)

```
month = [['Jan', 'Feb', 'March'],
         ['April', 'May', 'June'],
         ['July', 'August', 'Sept'],
         ['Oct', 'Nov', 'Dec']]
```

How would you use indexing to access the value August?

Answer: _____

Question 27 (6 points)

Match the following definitions to the correct example.

- Defines a TUPLE?
- Defines a DICTIONARY?
- Defines a LIST?

Examples to choose from:

```
["grape", "apple", "orange"]
{"name": "grape", "color": "green"}
("grape", "apple", "orange")
```

Question 28 (3 points)

Match the code to the expected results (True, False or Error).

```
pets = {"cat":15, "dog":6, "hamster":13}
print("dog" in pets)
```

```
pets = {"cat":15, "dog":8, "hamster":13}
print(13 in pets)
```

```
pets = {"cat":15, "dog":8, "hamster":13}
print(pets["dog"])
```

Question 29 (3 points)

Which is true?

A. Tuples are mutable while lists are immutable

B. Tuples are immutable while lists are mutable

C. Both tuples and lists are mutable

D. Both tuples and lists are immutable

Question 30 (3 points)

Match the code to the expected result (give the value of **numbers** after each).

```
numbers = [1, 2, 3, 4, 5, 6, 7]
numbers.pop(2)
```

```
numbers = [1, 2, 3, 4, 5, 6, 7]
numbers.pop()
```

Question 31 (3 points)

Which of the following are true of Python Lists?

- **A.** All answers are true
- **B.** A list may contain any type of element except another list
- **C.** All elements in a list must be of the same type
- **D.** An element may appear in a list more than once

Question 32 (3 points)

Which of the following are true of Python Dictionaries?

- **A.** Items are accessed by their position in a dictionary
- **B.** All answers are true
- **C.** Dictionaries are immutable
- **D.** Items are accessed by their key